

Wireless Network Traffic Forecasting Competition—Data Description

1. Dataset Overview

This competition provides **uplink** network traffic time series from **217 anonymized wireless network cells (Cells)**. Traffic is provided as the **sum** within each 5-minute window, with a sampling interval of **5 minutes**. The data continuously cover **7 days + 5 minutes**, and each cell contains **2,017** time points.

Each wireless network cell corresponds to an independent traffic time series. A full day contains:

$$24 \times 60 \div 5 = 288$$

time points. Therefore, the observation length for each cell in this dataset is:

$$288 \times 7 + 1 = 2,017$$

The time range is from 2024-11-30 23:00:00 to 2024-12-07 23:00:00, inclusive. The two endpoints are exactly 7 days apart; including both endpoints gives 2,017 time points per cell, corresponding to a total coverage of 7 days + 5 minutes.

The complete dataset contains **217 wireless network traffic time series**. Every cell has a record at every time point, for a total of:

$$217 \times 2,017 = 437,689$$

observations. The dataset captures intraday cycles, traffic peaks and troughs, and scale differences across cells, supporting wireless network traffic forecasting and network resource management.

2. Basic Dataset Information

Attribute	Description
Number of wireless network cells	217
Time points per cell	2,017
Dataset duration	7 days + 5 minutes
Temporal resolution	5 minutes
Samples per hour	12
Samples per day	288
Total observations	437,689
Missing records	0
Start time	2024-11-30 23:00:00 (Saturday)
End time	2024-12-07 23:00:00 (Saturday)
Time zone	Based on the recorded collection time; no separate time zone is specified
Original matrix shape	(T=2017, N=217), float32; values are integers in practice

Each record represents the sum of uplink traffic for the corresponding cell within a **5-minute aggregation window**.

The **timestamp** indicates the **start time of the aggregation window**. A timestamp t corresponds to the interval: $[t, t + 5 \text{ min})$

For example, 2024-11-30 23:00:00 corresponds to $[2024-11-30 \text{ 23:00:00}, 2024-11-30 \text{ 23:05:00})$.

The timestamps follow the actual collection calendar, covering Saturday, November 30, 2024 at 23:00 through Saturday, December 7, 2024 at 23:00. Participants may use timestamp-derived features such as day of week and weekend indicators. Because each day of the week appears only once, weekday effects should not be overfit.

3. Network Traffic Definition

The target variable **traffic** represents the **summed uplink traffic count** for the corresponding wireless network cell within a specified 5-minute window.

Attribute	Description
Metric name	Uplink Traffic
Aggregation window	5 minutes

Attribute	Description
Metric meaning	Total uplink traffic for the cell within the window
Aggregation method	Sum of uplink traffic within each 5-minute window
Data unit	Anonymized raw count unit
Traffic direction	Uplink traffic
Numeric transformation	No normalization, standardization, or logarithmic transformation
Value type	Non-negative integer; minimum 1 and maximum 28,952,520 in the full dataset

Observed distribution across all 437,689 records:

Statistic	Value
Minimum	1
1% quantile	8,914
25% quantile	174,401
Median	470,487
Mean	915,158
75% quantile	1,076,228
99% quantile	7,327,659
Maximum	28,952,520
Proportion with traffic $\geq 10^6$	27.09%

Mean traffic levels vary substantially across cells: the median of the per-cell means is approximately 5.80×10^5 , with the lowest at about 3.96×10^4 and the highest at about 7.92×10^6 .

4. Data Entities

The dataset contains **217 wireless network cells**, corresponding one-to-one with the column order in the original matrix.

Each cell is assigned a unique anonymized identifier, **cell_id**:

cell_id	Original column index	Meaning
Cell_001	0	First anonymized wireless network cell
Cell_002	1	Second anonymized wireless network cell
...	...	...
Cell_217	216	217th anonymized wireless network cell

To protect information about the actual communications network, the true cell names, physical locations, base-station identifiers, and any other information that could identify real network locations have been **anonymized and are not disclosed to participants**.

A given **cell_id** always refers to the same wireless network cell, so historical observations from the same cell may be used for modeling.

5. Data Fields

The data use a **wide-table format**: each row is a timestamp and each cell is represented by one column.

Field	Type	Description
timestamp	datetime	Start time of the current 5-minute aggregation window, in YYYY-MM-DD HH:MM:SS format; sorted in ascending time order
Cell_001 ... Cell_217	integer	Summed uplink traffic count for the corresponding cell within the time window

Example:

```
timestamp,Cell_001,Cell_002,...,Cell_217
2024-11-30 23:00:00,679910,184242,...,<value>
2024-11-30 23:05:00,588123,<value>,...,<value>
2024-11-30 23:10:00,738886,<value>,...,<value>
```

Reading a column from top to bottom gives the complete traffic time series for that cell. Reading a row from left to right gives the cross-sectional traffic values of all cells at the same timestamp.

6. Data Files

The following files are provided to participants.

6.1 train.csv

The training data contain historical traffic that may be used for modeling. Columns are `timestamp` and `Cell_001–Cell_217`.

Item	Value
Time range	2024-11-30 23:00:00 – 2024-12-06 22:55:00
Rows (time points)	1,728 (exactly 6 days)
Number of cell columns	217
Traffic observations	374,976

6.2 test.csv

The test data provide the timestamps to be predicted and the same cell columns as the training set. The cell columns are empty and contain **no** ground-truth traffic values.

Item	Value
Time range	2024-12-06 23:00:00 – 2024-12-07 23:00:00
Rows (time points)	289 (1 day + 5 minutes)
Number of cell columns	217
Entries to predict	62,713

The ground-truth traffic for the test segment is used by the competition platform for scoring and is not disclosed to participants.

6.3 sample_submission.csv

The sample submission has the same format as the scoring file: the first column is `timestamp`, followed by predicted values for `Cell_001–Cell_217`.

```
timestamp,Cell_001,Cell_002,...,Cell_217
2024-12-06 23:00:00,<prediction>,<prediction>,...,<prediction>
2024-12-06 23:05:00,<prediction>,<prediction>,...,<prediction>
...
```

- The submission must include every timestamp in `test.csv` and all 217 cell columns, with no missing or additional rows or columns.
- The row order and column order should preferably remain identical to `test.csv`.
- Each cell column must contain numeric values; predictions may be floating-point numbers.
- Values in the sample file are generated by a **seasonal naive baseline** (using traffic from the same time on the final training day) and are provided only to demonstrate the required format; they are not reference answers.

7. Training/Test Split

The dataset is split **chronologically**; no random sampling is performed across cells or time points.

Training Time Range

```
2024-11-30 23:00:00 -- 2024-12-06 22:55:00
(windows 0 through 1727, 1,728 points / 6 days in total)
```

Test Time Range

2024-12-06 23:00:00 -- 2024-12-07 23:00:00

(windows 1728 through 2016, 289 points / 24 hours 5 minutes in total)

Models may use only information that has already been provided before each test time. Test-segment labels are unavailable before evaluation, making this a **multi-series, multi-step direct forecasting** task: forecast the next 289 consecutive 5-minute points for all 217 cells.

Permitted information includes:

- Historical traffic from the same cell;
- Timestamp-derived features such as temporal cycles and intraday position;
- Correlations across different cells;
- Other features constructed solely from the publicly provided training data.

The ground-truth **traffic** values in the test segment and any non-public external operational data are not permitted.

8. Data Completeness and Anomalies

8.1 Missing Values

All 217 cells have records at all 2,017 time points. There are **no** NaN values, infinite values, or missing timestamps. The time axis is a strictly regular 5-minute grid and requires no gap filling.

8.2 Zero Values

There are **no** observations with **traffic** = 0 in the released dataset. All observations satisfy **traffic** ≥ 1 .

If a participant obtains zeros during intermediate processing, they should not be interpreted as missing values in the source data.

8.3 Outliers and Level Shifts

Traffic contains short-term peaks, with a maximum of approximately 2.90×10^7 . Such bursts of high traffic may reflect genuine changes in network activity and should not automatically be treated as data-entry errors.

In addition, a small number of cells exhibit clear level shifts from one day to the next (the maximum day-to-day change in the log daily mean can exceed 1.0). These cells are retained in the full dataset. Participants may choose whether to apply anomaly detection, segmentation, or robust processing.

8.4 Intraday Cycle

According to the collection clock, average traffic across the full dataset is lowest at approximately 04:00 and highest at approximately 21:00, showing a typical intraday peak–trough pattern. The strength and phase of the cycle are not identical across cells.

9. Data Anonymization and Privacy Protection

The dataset has been anonymized:

- It contains no user identity information;
- True geographic locations, base-station names, and other information that could identify network infrastructure are not provided;
- **cell_id** is used only to distinguish cells and carries no geographic or business meaning;
- Timestamps retain the actual collection calendar, but cell locations are not provided, so specific network infrastructure cannot be identified from the timestamps.

Traffic values retain the raw counts from the source data and have **not** undergone any additional normalization or scaling. The values cannot be directly converted into real physical traffic units.

10. Dataset Characteristics

This dataset is a **high-frequency, multi-series wireless network traffic time-series dataset**.

10.1 High Temporal Resolution

The temporal resolution is 5 minutes, giving 288 points per cell per day and capturing fine-grained changes in network activity over time.

10.2 Multiple Time Series

The dataset covers 217 cells simultaneously. Traffic scale, periodicity, and volatility vary substantially across cells.

10.3 Intraday Periodicity

Traffic follows the clock, with troughs in the early morning and peaks in the evening. The primary period is 288 time points (1 day).

10.4 Cross-Cell Heterogeneity

Mean traffic levels can differ by more than two orders of magnitude across cells, so a single set of hyperparameters may not be optimal for every cell.

10.5 Potential Cross-Cell Correlation

Different cells may change synchronously during the same time periods and may therefore be modeled jointly.

10.6 Short History

Each cell has only about 7 days of history. Long-period effects such as weekly patterns cannot be estimated reliably from the data, so models should rely primarily on intraday periodicity, recent traffic levels, and cross-cell information.

11. Submission Requirements

Prediction files submitted by participants must satisfy the following requirements:

1. The first column must be `timestamp`, followed by `Cell_001` through `Cell_217`; column names and order must match `test.csv`.
2. The set of rows must correspond one-to-one with the `timestamp` values in `test.csv`, with no missing or additional rows.
3. A prediction must be provided for every timestamp in every cell column.
4. Prediction values must be numeric.

Non-negative predictions are recommended. If any non-numeric or missing entry is found during evaluation, the submission will be considered invalid.

12. Evaluation Metrics

This competition uses **5 official metrics**. **Each metric is ranked independently and awards one winner**. Participants need to submit only one prediction file; the platform computes all metrics separately on the same set of predictions.

Let all test-set entries be $\{(y_i, \hat{y}_i)\}_{i=1}^n$, where $n = 62,713$. Masked MAPE is calculated only on entries whose ground-truth values meet the specified threshold:

Track	Metric	Definition	Evaluation Scope
1	MAE	$\frac{1}{n} \sum_i \hat{y}_i - y_i $	All test-set entries
2	RMSE	$\sqrt{\frac{1}{n} \sum_i (\hat{y}_i - y_i)^2}$	All test-set entries
3	MAPE	$\frac{100}{n} \sum_i \hat{y}_i - y_i / y_i$	All test-set entries
4	Masked MAPE @ 3e5	$\frac{100}{n} \sum_i \hat{y}_i - y_i / y_i$	$y_i \geq 3 \times 10^5$ (approximately 39,144 test-set entries, 62.4%)
5	Masked MAPE @ 1e6	$\frac{100}{n} \sum_i \hat{y}_i - y_i / y_i$	$y_i \geq 10^6$ (approximately 17,132 test-set entries, 27.3%)

For all five metrics, **lower is better**. Each track has an independent winner: the participant with the lowest value for that track's metric wins. If two valid submissions have exactly the same value on the same track, the earlier valid submission wins the tie. The winners of different tracks may be the same team or different teams.

The mask for Masked MAPE is determined solely by the **ground-truth** value y_i , independently of the prediction. Entries outside the threshold are excluded from the average for that track.

13. Dataset Size Summary

Metric	Value
Number of wireless network cells	217
Temporal resolution	5 minutes
Samples per hour	12
Samples per day	288
Dataset duration	7 days + 5 minutes
Observations per cell	2,017
Number of time series	217
Total observations	437,689
Training table size	1,728 rows $\times$ 217 cells
Test table size	289 rows $\times$ 217 cells
Training traffic observations	374,976
Test entries to predict	62,713
Forecast steps	289